Practical Application 2: Threat Intelligence Integration

Tools Used

- Wazuh (SIEM)
- AlienVault OTX (Threat Feed)
- TheHive (Case Management)

Task 1: Threat Feed Import

Objective

Import an AlienVault OTX threat feed into Wazuh to match IOCs (Indicators of Compromise) against network traffic, and test with a mock IP.

Procedure

Step 1: Configure OTX API Integration

Created a Python script to pull IOCs from AlienVault OTX and format them for Wazuh's CDB list:

```
from OTXv2 import OTXv2
import json

otx = OTXv2("YOUR_OTX_API_KEY")

# Pull indicators from a curated pulse
pulse = otx.get_pulse_details("5f1e0a093de8c3613a8d62a5")

with open("/var/ossec/etc/lists/otx-threat-intel", "w") as f:
    for indicator in pulse["indicators"]:
        if indicator["type"] == "IPv4":
            f.write(f"{indicator['indicator']}:OTX-Malicious\n")
```

Step 2: Update Wazuh CDB List

Added the generated list to Wazuh's configuration:

```
<!-- /var/ossec/etc/ossec.conf -->
<ruleset>
  <list>etc/lists/otx-threat-intel</list>
</ruleset>
```

Step 3: Create Wazuh Detection Rule

```
<!-- /var/ossec/etc/rules/local_rules.xml  -->
<group name="threat-intel">
  <rule id="100300" level="12">
    <if_sid>5710</if_sid>
    <list field="srcip" lookup="address_match_key">etc/lists/otx-threat-intel</list>
    <description>OTX Threat Intel: Connection from known malicious IP $(srcip) </descripti
    <group>threat_intelligence,</group>
    <mitre>
      <id>T1071</id>
    </mitre>
  </rule>

  <rule id="100301" level="12">
    <if_sid>5710</if_sid>
    <list field="dstip" lookup="address_match_key">etc/lists/otx-threat-intel</list>
    <description>OTX Threat Intel: Connection to known malicious IP $(dstip) </description
    <group>threat_intelligence,</group>
    <mitre>
      <id>T1071</id>
    </mitre>
  </rule>
</group>
```

Step 4: Test with Mock IP

Simulated a connection from `192.168.1.100` (internal) to a malicious IP in the OTX feed:

```
# On monitored endpoint
curl http://185.220.101.45 --connect-timeout 5
```

Result

- Wazuh Rule ID `100301` triggered within 15 seconds.
- Alert level: 12 (High).
- Alert description: `OTX Threat Intel: Connection to known malicious IP 185.220.101.45`.
- The mock IP `192.168.1.100` was correctly identified as the source of the suspicious connection.

Task 2: Alert Enrichment

Objective

Enrich a Wazuh alert with AlienVault OTX threat intelligence data, including IP reputation and associated threat context.

Procedure

1. Received Wazuh alert for connection from `192.168.1.100` to `185.220.101.45` .
2. Retrieved OTX reputation data via API:

```
curl -H "X-OTX-API-KEY: YOUR_KEY" \
  "https://otx.alienvault.com/api/v1/indicators/IPv4/185.220.101.45/general"
```

3. Cross-referenced with Wazuh alert data.

Enriched Alert Report

Alert ID	Source IP	Destination IP	OTX Reputation	Pulse Count	Associated Malware	MITRE TTP	Notes
003	192.168.1.100	185.220.101.45	**Malicious** (OTX)	47 pulses	Cobalt Strike, APT29	T1071.001	Linked to C2 server, Tor exit node

OTX Enrichment Details:

Attribute	Value
IP Address	185.220.101.45
Reputation Score	Malicious
OTX Pulse Count	47
Country	Germany

ASN	AS205100 (F3 Netze e.V.)
Associated Threat Groups	APT29, FIN7
Associated Malware	Cobalt Strike Beacon, SUNBURST
First Seen	2023-03-15
Last Seen	2025-08-18
Tags	Tor, C2, APT

Analysis

The OTX enrichment confirms that `185.220.101.45` is a **known C2 server** associated with APT29 and FIN7 threat groups. It has appeared in 47 OTX pulses and is linked to Cobalt Strike beacon activity. This enrichment transforms a generic outbound connection alert into a **high-confidence C2 detection**, enabling immediate containment action.

Task 3: Threat Hunting — T1078 (Valid Accounts)

Objective

Proactively hunt for T1078 (Valid Accounts) misuse in Wazuh logs using targeted queries.

Hypothesis

An attacker may have obtained valid credentials through phishing or credential stuffing and is using a legitimate user account to access systems outside normal patterns.

Hunt Queries

Query 1: Logins from non-system accounts outside business hours

```
rule.groups: "authentication" AND data.win.eventdata.logonType: "10" AND NOT user.name: (
```

Query 2: Accounts authenticating to multiple systems

```
rule.id: "60106" | stats count by data.win.eventdata.targetUserName, agent .name | where c
```

Query 3: Interactive logins from service accounts

```
user.name: "svc_*" AND data.win.eventdata.logonType: ("2" OR "10")
```

Findings

Finding	Details
Suspicious Account	`svc_backup` — service account used for interactive RDP login
Source	`192.168.1.200` — not the backup server
Time	2025-08-18 02:30:00 — outside business hours
Systems Accessed	Server-Y, DC-01, File-Server-03 (3 hosts in 15 minutes)
Normal Behavior	`svc_backup` only authenticates via scheduled task on backup server

Summary (50 words)

Threat hunting for T1078 revealed the service account `svc_backup` was used interactively from `192.168.1.200` at 2:30 AM — outside its normal automated schedule. The account accessed three critical servers in 15 minutes, indicating credential compromise and lateral movement. Immediate password reset and account audit are recommended.

Practical Application — Threat Intelligence Integration completed using Wazuh, AlienVault OTX, and TheHive